Sheet 5

Q1.

a). Bob starts off with a uniform prior about A's card X, and upon being called to act updates his beliefs to some posterior (which depends on A's eq* strategy). It's BR for Bob to call iff

$$P(X < y \mid \text{raise}) \cdot 3 + P(X > y \mid \text{raise}) \cdot -1 \geq 0$$

$$\Leftrightarrow \quad P(X < y \mid \text{raise}) \geq 1/4.$$

Great

But $P(X < y \mid \text{raise})$ is simply the posterior CDF's value at Y, which monotonically increases in Y. So there's some threshold $Y = c$ at which the ~~strategy~~ BR switches from fold to call.

$\Rightarrow$ Bob's $\mathbb{E}$ payoff from calling vs folding increases in Y, so must be threshold strategy.

Ann doesn't know ~~X~~ Y, but $P(Y \leq c) = c$, by properties of $U[0,1]$.

b) Ann's payoff from checking is zero (take buy-in as sunk; she can't win)

From raising, payoffs are $+2$ w.p. of Bob folding; -1 w.p. of Bob ~~a~~ calling (since in comparison she's guaranteed to lose).

So it's better to raise iff $2c + -1(1-c) > 0 \Leftrightarrow c > 1/3$.

This demonstrates how bluffing might arise. When c is sufficiently large, i.e. B needs to observe a ~~large~~ high value of Y to want to call, then A might raise even with a terrible card, because of the chance that B doesn't call and instead folds being more favourable than A accepting defeat from checking.

(Presumably very large values of c will not give rise to PBE, because then B won't update at all from A raising - so posterior will simply be uniform prior, i.e. "bluffing" isn't possible from A.)

c) Bluff case: we require

$$2 \cdot P(A \text{ wins from checking}) \leq 2 \cdot P(B \text{ folds}) + -1 \cdot P(B \text{ calls}, A \text{ loses}) + 3 \cdot P(B \text{ calls}, A \text{ wins})$$

i.e. $2 \cdot P(Y < x) \le 2c + -1 \cdot (1-c) \cdot P(Y > x \mid B \text{ calls})$
$+ 3 \cdot (1-c) \cdot P(Y < x \mid B \text{ calls})$

but by hypothesis B calls iff $Y > c$, so conditional distribution is $\sim U[c, 1]$. And moreover, $a < c$. So in fact, A <u>never</u> wins if B calls, if she bluffed.

$\iff 2 \cdot P(Y < x) \le 2c + (c-1)$, with $P(Y < x) = x$.

$2x \le 3c - 1$ and at the threshold $x = a$

$\underline{2a = 3c - 1}$ by indifference

$P(Y < x \mid B \text{ calls})$ has $Y \in [c, 1]$

"True raise" case: here ~~A wins for sure whenever B calls~~
We require, similarly to before:
$2 \cdot P(Y < x) \le 2c + 3(1-c) \cdot x - 1(1-c) \cdot (1-x)$
So at threshold $x = b$
$2b = 2c + (1-c) \cdot (3b - 1 + b) = 2c + 4b - 4bc - 1 + c$

Bob's indifference: $P(X < y \mid \text{raise}) = 1/4$, from part a)

"True raise" case: here A might or might not win, after raising + calling. If B calls we know $\sim U[c, 1]$.

To prefer to raise,

[I feel like I'm not getting anywhere with this + am just going to leave it]

"True raise": $x \geq b > c$.

- ~~E~~ Payoff from folding is as before, $2 \cdot P(Y < x) = 2x$
- And also as before, A gets 2 from raise, fold w.p. c.
- But if challenged post-raise, A may or may not win. As noted, given that B calls, conditionally $Y \sim U[c, 1]$.
  - So, $P(\underbrace{Y < x}_{A \text{ winning}} \mid B \text{ calls}) = \dfrac{P(c < Y < x)}{P(Y > c)} = \dfrac{x - c}{1 - c}$

Thus at point of indifference $x = b$,

$$2b = 2c + -1 \cdot \frac{1-b}{\cancel{1-c}} \cdot \cancel{(1-c)} + 3\,\frac{b-c}{\cancel{1-c}} \cdot \cancel{(1-c)}$$

$$P(A \text{ losing} \mid B \text{ calls}) = \frac{P(Y > x)}{P(Y > c)} = \frac{1-x}{1-c}$$

$\therefore\ 2b = 2c - 1 + b + 3b - 3c$

$\Rightarrow\ \underline{2b = c + 1}$

Bob's indifference: as in part (a), $P(X < c \mid A \text{ raises}) = 1/4$ is the condition for our threshold $y = c$.

~~$P(X < y \mid A \text{ raises}) = 2P(X < y \mid A \text{ bluffs}) = \ldots$~~

If A does a bluff, then we know $X \leq a < c$ so B wins for sure. Conversely, if A does a true raise then with $Y = c < b \leq X$, B loses for sure. i.e. only the bluff part is relevant.

$P(A \text{ bluffs})$ is simply $P(X \leq a) = a$; $P(A \text{ true-raises}) = P(X \geq b) = 1 - b$;

$\therefore\ P(\text{raises}) = a + 1 - b$.

So $\dfrac{a}{a+1-b} = \dfrac{1}{4} \Rightarrow \underline{3a = 1 - b}$

Wow! Nice!!!

(1) $2a = 3c - 1$
(2) $2b = c + 1$
(3) $3a = 1 - b$.

$b = 1 - 3a$, $2 - 6a = c + 1$, $c = 1 - 6a$.

$2a = 3 - 18a - 1 \Rightarrow$ $\boxed{a = 1/10,\ b = 7/10,\ c = 4/10}$

As required, $a < c < b$, $c > 1/3$.

Bluffing isn't really "dishonest" in that B can compute for themself that there's PBE where A will raise with bad cards. At no point does A claim that she raises only when her cards are good — and indeed it's common knowledge she doesn't.

d) It seems better to be Ann, since she gets to "force the hand" of Bob more, e.g. by deciding whether or not he's even called upon to act.

→ Yes. Ann's outside option has zero expected payoff. In eqm she does better than that. The game is zero-sum so Bob does worse.

2.

1, 1	b, a
a, b	0, 0

$a \geq 1, \quad b < 0, \quad a + b \leq 1$

$p + (1-p) b \geq 0.$

1, 1	$-\frac{1}{2}$, $\underline{\frac{3}{2}}$
$\underline{\frac{3}{2}}$, $-\frac{1}{2}$	$\underline{0}$, $\underline{0}$

a) $\underline{t=2}$

For each player, D is strictly dominant. So rational-Ann (A_r) and Bob must both play D. ~~[illegible]~~

$\underline{t=1}$

- Tit-for-Tat-Ann (A_t) plays C for sure.
- A_r knows that whatever she plays now, B will play D next round. So she can consider only this round as relevant, where it's again strictly dominant to play D.

- If faced with A_r, it's BR for B to play D, clearly.
- If faced with A_t, ~~[illegible]~~ B should play C in round 1, to get payoffs associated with (C, C) + (C, D) rather than (C, D) + (D, D), given that $\underset{(C,C)}{1} > \underset{(D,D)}{0}$.

But of course B doesn't know which type he's facing, just has a prior that it's A_r w.p. $1-p$, A_t w.p. p.

$v_B(DD) = 0 \cdot (1-p) + a \cdot p + 0 \cdot (1-p) + 0 \cdot p$

$v_B(CD) = b \cdot (1-p) + 1 \cdot p + 0 \cdot (1-p) + a \cdot p$

(In both cases, we defect in $t=2$, but A_t's response varies)

$\therefore\ v_B(DD) = ap \quad ; \quad v_B(CD) = b - bp + p + ap$

So Bob's BR is to C in $t=1$ whenever

$ap \le b - bp + p + ap \implies p \geq \frac{b}{b-1}$ since $b-1<0$.

and D otherwise, but $\iff p + (1-p)b \geq 0$ assumption.

For a PBE we need assessments, i.e. strategies and beliefs.

- A_r plays (D, D); believes B is rational w.p. 1 everywhere.
- A_t plays (C, B_{t1}); — " —
- B plays (CD); starts with prior of $\mu_0 = p$ of A TFT; updates to $\mu_1 = 1$ if observes C and $\mu_1 = 0$ otherwise.

[Hmm, are the beliefs meant to characterise types or expected actions/strategies? Here they do coincide due to separation but am not sure how you're meant to write them out for PBE.]

The beliefs will be about actions, but those are determined by types...
For the PBE, you just need to write what each player's beliefs are at each of their info sets.

Yes, that's right. This is why the question specifies exactly what the initial actions are.

b) Backward induction would find the same eq* path as described above, just now for $t=3$ and $t=2$. So we just need to consider what happens in the first round. (Note that the one thing that might've changed is A_t's action, but since in the candidate PBE B plays C at $t=1$, A_t will play C for sure at $t=2$, the old $t=1$, as required.)

[If there weren't pooling, or if now B played D to start, then would we have a different continuation to the PBE found in a), since the initial conditions changed?]

We need to show (i) that it's optimal for A_r to play C at $t=1$, (ii) that B will also do so, (iii) that B's beliefs at $t=2$ are consistent with the continuation described in a).

(iii) is trivial because in this candidate eq* there's pooling at $t=1$ so B makes no update at all. So (on-path) at $t=2$ B's posterior $\mu_2 = p$ is the same as his prior, and we established already in a) that this is compatible with our continuation.

find BR for B. ~~know A_t plays D at t=2,3~~

~~[illegible]~~ We know A_r will play D at $t=3$ even off-path, and after B plays D at $t=1$, A_t plays D at $t=2$. So A_r would also play $D_{t=2}$ since no reputational value from C and D strictly dominant in stage-game. All to say, if B plays D at $t=1$, A_r and A_t will play D every subseq. round, (and B should continue with D)

$\therefore v_B(CCD) \geq v_B(DDD)$ is required for (ii).

$v_B(CCD) = 1 + p + (1-p)b + ap$ as guaranteed 1, then $v_B(CD)$ from (a).

What about $V_B(CCD)$ vs. $V_B(DCD)$?

$v_B(DDD) = a + 0 + 0 = a.$

$$\Rightarrow \underbrace{p + (1-p)b}_{\geq 0 \text{ by prev. assumption}} + \underbrace{ap - a \geq -1}_{\text{expecting this}} \text{ required}$$

$\iff (1-p)a \leq 1$, which is new assumption.

So (ii) holds.

(i) demands that $v_{A_r}(CDD) \geq v_{A_r}(DDD)$, given B plays as above.

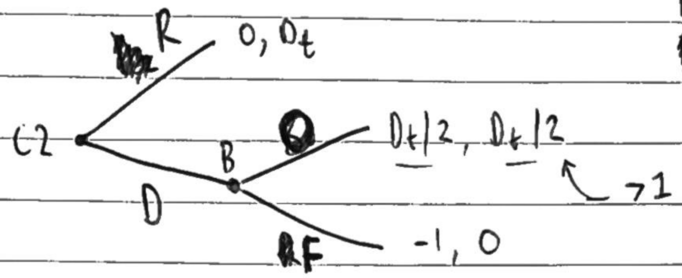
So we need $1 + \underset{\substack{\uparrow \\ \text{when B plays} \\ \text{C but she Ds}}}{a} + 0 \geq a + 0 + 0$ which is trivially true.

For A_r, it makes sense to C at $t=1$ so that she can pass as A_t, i.e. develop a reputation for retaliation, and then reap benefits by defecting on B later.

This corresponds to the general ~~f~~ repeated PD result that we need the short-term gains from D to be outweighed by the long-term costs due to retaliational punishment, to sustain (C,C).

R = repay   D = default
F = refuse   O = bailout

3. a)



By backward induction, SPE is (D, O)

b) The first period is just a duplicate of second period; no "reputation" or similar relevant since common knowledge of rationality. So the SPE will be (D, D, O O)

c) We have that normal bank (B_n) refuses to bailout C_1 w.p. $q \in (0,1)$.

So for C_2, $P(B \text{ is tough} \mid C_1 \text{ refused}) = \dfrac{1 \cdot 1/2}{1/2\,(q+1)} = \dfrac{1}{q+1}$

since $P(\text{refused}) = \frac{1}{2}\left[P(\text{refused} \mid B_n) + P(\text{refused} \mid B_t)\right]$,
$P(\text{ref} \mid \text{tough}) = 1$ and $P(\text{tough}) = 1/2$.

For C_2 to be indifferent between R and D given its posterior, we require $v_{C_2}(R \mid C_1 \text{ ref}) = v_{C_2}(D \mid C_1 \text{ ref})$. And this is necessary for them to mix as described.

$v_{C_2}(R \mid C_1 \text{ ref}) = 0$ for certain, unchanged by observation.
$v_{C_2}(D \mid C_1 \text{ ref}) = D_2/2 \cdot \left(1 - \frac{1}{q+1}\right) - 1 \cdot \frac{1}{q+1}$

Since, given it's final period, B_n would surely respond to D with O, as it's strictly better for B_n.

$\Rightarrow \dfrac{1}{q+1} = \dfrac{D_2}{2} \cdot \left(1 - \dfrac{1}{q+1}\right)$, $\dfrac{1}{q+1}\left(1 + \dfrac{D_2}{2}\right) = \dfrac{D_2}{2}$

so $q = \dfrac{1 + D_2/2}{D_2/2} - 1 = \dfrac{2}{D_2} \in (0,1)$ since $D_2 > 2$.

Good! What about what C_2 does after "default, bail" and after "pay"?

For this to be PBE, it needs to be rational for B_n to mix between O and F when responding to C_1, i.e. indifference must hold.

$$v_{B_n}(O\,O) = D_1/2 + D_2/2$$

since then C_2 knows B is normal, so C_2 plays D and B BRs with O again.

$$v_{B_n}(F, O) = 0 + p \cdot D_2 + (1-p) \cdot D_2/2 \quad \text{with } p = D_1/D_2$$

if C_2 does D, still BR for B_n to bailout.

$\therefore$ we require $D_1/2 + D_2/2 = \cancel{D_1/D_2 \cdot D_2} + \underbrace{(1 - D_1/D_2) \cdot D_2/2}_{D_2/2 - D_1/2}$

$$= D_1 + D_2/2 - D_1/2 = \frac{D_1 + D_2}{2}$$

i.e. indifference holds with no further restrictions needed.

For C_1, we have $v_{C_1}(R) = 0$ and $v_{C_1}(D) = -\frac{1}{2} + \frac{1}{2}\left[-q + (1-q) \cdot \frac{D_1}{2}\right]$ against B_t ... B_n

So it would pay up whenever

Nice

$$(1-q)\frac{D_1}{2} - q < 1 \iff (1 - 2/D_2)\frac{D_1}{2} - 2/D_2 \leq 1$$

At $D_1 = 4$, $D_2 = 6$ LHS is $(1 - 1/3) \cdot 2 - 1/3 = 1$ so condition binds, when larger values of D_2 it'll be slack.

PBE characterised by:

- B_t always refuses to bailout.
- B_n bails out w.p. $q = 2/D_2$ at $t = 1$ and for sure at $t = 2$
- C_1 has prior $\mu = P(\text{tough}) = 1/2$ and repays for sure provided that $(1/2 - 1/D_2)D_1 - 2/D_2 \leq 1$
- C_2 has on-path belief $\mu = 1/2$ since no update, but if C_1 defaults then following O we have $\mu = 0$, following F let $\mu = 1/q+1$ with $q = 2/D_2$. [But other values of μ also would work?]

[Hmm, but we could've imposed any beliefs off-path for PBE, so why did we need to calculate Bayesian-updated one, to pin down q?]

The history is reached with positive probability in eqm, so beliefs must be updated via Bayes rule there.

Strategy for c_2 is to repay w.p. $p = D_1/D_2$ when ~~...~~ when $\mu = 1/q+1$ (i.e. after c_1 default and no bailout); D for sure when $\mu = 0$, and also D for sure on-path when $\mu = 1/2$ since $\frac{1}{2} \cdot \frac{D_2}{2} - \frac{1}{2} > 0$ with $D_2 > 2$.

d) Note that on-path c_1 might pay up given conditions on D_1, D_2 but c_2 certainly won't. So the bank does benefit from "reputation" at $t = 1$ by getting D_1 rather than $D_1/2$, but not at $t=2$, rather counter-intuitively.

If $D_1 > D_2$ then the stakes for $t=2$ are higher than first period. Mechanically, $p = D_1/D_2$ will not be a valid probability so it's not going to be PBE. Intuitively, it's not worth it for a normal bank to pretend to be tough at $t=1$ in order to try and make c_2 pay up at $t=2$, so c_1 can safely default knowing that the rational bank would bail them out, and c_2 would know for sure either way whether bank is normal or tough.

So EV-positive to D